AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM240
Introduction to Computer Graphics

Midterm Examination (MTA)- Makeup
Spring Semester 2025/2026

Date: 29 /Apr./ 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 90

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Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- *Be sure you write your* **name and ID** on the External Answer booklet.
- ***Calculators*** *are allowed.*

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 .....space is attached to a device.

A Device B Object
C view D None of the above

2 The function intersect is a member of the class .....

A Graphics B Area
C Shape D Graphics2D

3 The top left corner of the ellipse defined by Ellipse2D.Float(20f,30f,100f,80f) is at .....

A (20,80) B (100,80)
C (20,30) D (80,70)

4 .....is a set theoretic function used to construct the union of two areas.

A interset B add
C union D exclusiveor

5 In the instruction Arc2D arc=new Arc2D.Float(20, 30, 100, 80, 45, 120) the starting angle of the arc is at 45 while the ending angle is at .....

A 30 B 80
C 120 D 165

6 The function intersect is a member of the class .....

A Area B Shape
C Graphics2D D Graphics

7 The .....transform can be used to convert circle to ellipse.

A Shearing B Translation
C Rotation D Scaling

8 *Win32 API graphics is an example of .....programming.*

A GKS B Hardware level
C Operating system D None of the above

9 .....*produces dynamic graphics contents and rendering.*

A Animation B Interaction
C Viewing D None of the above

10 *The function scale() belonging to the class Graphics can be considered as .....function.*

A Object transformation B Device transformation
C Viewing transformation D None of the above

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

Draft the output of this code segment:

```
Ellipse2D e=new Ellipse2D.Double(0,0,200,200);
AffineTransform af=new AffineTransform();
float [] da={3,3};
for (int i=0;i<5;i++)
{
  af.translate(100, 100);
  af.scale(0.8f, 0.8f);
  af.translate(-100, -100);
    if(i%2==0)
    g2.setStroke(new
BasicStroke(1,BasicStroke.CAP_ROUND,BasicStroke.JOIN_BEVEL,0,da,0));
  else
    g2.setStroke(new BasicStroke(2));
  g2.draw(af.createTransformedShape(e));

}
```

Question 3: (15 marks)

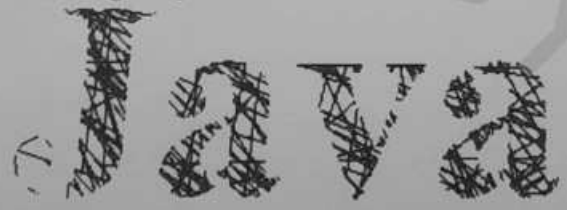
Draft the output of the following java code:

```
Area plate = new Area(new Rectangle2D.Double(50, 50, 400, 200));

  for (int y = 80; y <= 220; y += 70) {
    for (int x = 100; x <= 400; x += 80) {
      Area hole = new Area(
        new Ellipse2D.Double(x - 15, y - 15, 30, 30)
      );
      plate.subtract(hole);
    }
  }
  g2.draw(plate);
```

Question 4: (15 marks)

Having the following shape:



Write a java code segment to plot it, assuming that the lines inside the shape are randomly positioned.

End of questions

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